

SOFTWARE REQUIREMENTS SPECIFICATION

Web Based ML Model Evaluator

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Author / SRN	Role
PES2UG24CS578	Project Author
PES2UG24CS592	Project Author
PES2UG24CS599	Project Author
PES2UG24CS604	Project Author

Document Purpose

This SRS defines the functional requirements, non-functional requirements, interfaces, security requirements, acceptance criteria, system models, and traceability requirements for a web application that evaluates built-in machine-learning algorithms on user-provided datasets.

Document Control

Revision History

Date	Version	Description	Authors
30-09-2026	1.0	Revised SRS: improved structure, concise requirements, updated authorship	PES2UG24CS578, 592, 599, 604

Approval

Role	Name / SRN	Signature	Date
Project Authors	PES2UG24CS578, PES2UG24CS592, PES2UG24CS599, PES2UG24CS604		
Instructor			

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1. Introduction

1.1 Purpose

This document specifies the requirements for a web-based machine-learning model evaluator. The system allows users to upload a dataset, configure an evaluation, select built-in ML algorithms, execute training and testing on the server, view evaluation metrics and visualizations, and export results.

1.2 Scope

- Dataset upload, validation, preview, and target/feature selection.
- Selection and configuration of supported ML algorithms.
- Server-side model training and evaluation.
- Display of standard classification metrics and visualizations.
- PDF and CSV result export.
- Authentication and evaluation-history storage for registered users.

Out of scope: user-supplied pretrained models, distributed/large-scale training, and production-grade MLOps functions such as model versioning and deployment pipelines.

1.3 Intended Users

- Students and analysts who need a quick way to evaluate and compare ML algorithms.
- Developers responsible for implementation and maintenance.
- QA evaluators, instructors, and teaching assistants responsible for verification.

1.4 Definitions and Acronyms

Term	Definition
ML	Machine Learning
CSV	Comma-Separated Values
API	Application Programming Interface
UI	User Interface
F1-score	Harmonic mean of precision and recall
ROC-AUC	Receiver Operating Characteristic — Area Under Curve
JWT	JSON Web Token
RTM	Requirements Traceability Matrix

2. Overall Description

2.1 Product Perspective

The product is a standalone web application consisting of a browser-based frontend, a backend service for dataset processing and model evaluation, and a database for user accounts and evaluation history. The system operates independently of external enterprise systems.

2.2 Major Functions

- User registration, login, and logout.
- CSV dataset upload, validation, parsing, and preview.
- Target-column and feature selection.
- Built-in algorithm selection and train-test split configuration.
- Server-side model training and evaluation.
- Metric calculation, visualization, and multi-algorithm comparison.
- PDF and CSV report export.
- Per-user evaluation history for registered users.
- Optional administrator view for usage statistics and logs.

2.3 User Roles

Role	Capabilities
Guest	Upload and evaluate datasets; view current results; cannot save history.
Registered User	All guest capabilities plus save, view, and delete personal evaluation history.
Admin (Optional)	View system-level usage statistics and logs.

2.4 Operating Environment

- Client: modern web browsers such as Chrome, Firefox, and Edge.
- Server: standard server/container environment with Python or Node.js runtime.
- ML framework: a standard ML library such as scikit-learn.
- Connectivity: Internet connection required; no offline mode.

2.5 Constraints

- Initial dataset format is CSV.
- Maximum dataset size is 10 MB.
- Only built-in algorithms are supported.
- Production deployment shall use HTTPS and secure password hashing.

3. External Interface Requirements

3.1 User Interface

The application shall provide a responsive dashboard with a clear workflow: **Upload** → **Configure** → **Evaluate** → **Results**. Result tables and charts shall use readable labels and accessible contrast.

3.2 Hardware Interface

No specialized hardware is required. The system shall run on standard desktop, laptop, or tablet devices with a supported browser.

3.3 Software Interfaces

- REST backend API using JSON over HTTPS.
- ML library interface for model training and evaluation.
- Database interface for user accounts and evaluation history.

3.4 Communication Interface

- HTTPS/TLS 1.2 or later for deployed client-server communication.
- JWT or secure session-based authentication for protected requests.
- Standard HTTP status codes with structured JSON error responses.

4. Functional Requirements

All functional requirements use the identifier format MLE-F-####.

4.1 Authentication

ID	Requirement
MLE-F-001	The system shall allow a new user to register with a unique email/username and password.
MLE-F-002	The system shall store passwords using secure one-way hashing; plaintext passwords shall not be stored.
MLE-F-003	The system shall allow registered users to log in using valid credentials.
MLE-F-004	The system shall reject invalid login credentials using a generic error message.
MLE-F-005	The system shall allow logged-in users to log out and invalidate the active session/token.

4.2 Dataset Upload and Validation

ID	Requirement
MLE-F-006	The system shall accept datasets in CSV format.
MLE-F-007	The system shall reject unsupported file formats with a clear error message.
MLE-F-008	The system shall reject files exceeding the configured maximum size.
MLE-F-009	The system shall parse a valid CSV and display its column names and a preview of the first N rows.

4.3 Dataset and Target Configuration

ID	Requirement
MLE-F-010	The system shall allow the user to select one target column.
MLE-F-011	The system shall allow the user to select or deselect feature columns.
MLE-F-012	The system shall detect and handle missing values using a defined strategy and inform the user of the strategy applied.

4.4 Algorithm Selection and Configuration

ID	Requirement
MLE-F-013	The system shall provide Logistic Regression, Decision Tree, Random Forest, SVM, and K-Nearest Neighbors as built-in algorithms.
MLE-F-014	The system shall allow one or more algorithms to be selected for a single evaluation run.
MLE-F-015	The system shall allow the user to configure a valid train-test split ratio.

4.5 Model Evaluation Engine

ID	Requirement
MLE-F-016	The system shall train each selected algorithm on the training split and evaluate it on the test split.
MLE-F-017	The system shall calculate accuracy, precision, recall, and F1-score for each applicable evaluation.
MLE-F-018	The system shall generate a confusion matrix for classification evaluations.
MLE-F-019	The system shall calculate ROC-AUC where applicable, including binary classification with probability output.

4.6 Results Visualization and Comparison

ID	Requirement
MLE-F-020	The system shall display charts comparing selected metrics across evaluated algorithms.
MLE-F-021	The system shall display each applicable confusion matrix as a heatmap or grid.
MLE-F-022	The system shall provide a comparison table that can be sorted by a selected metric.

4.7 Report Export

ID	Requirement
MLE-F-023	The system shall export evaluation results as a PDF report.
MLE-F-024	The system shall export raw metric results as a CSV file.

4.8 Evaluation History

ID	Requirement
MLE-F-025	The system shall save evaluation-run metadata, including dataset name, selected algorithms, metrics, and timestamp, for logged-in users.
MLE-F-026	The system shall allow logged-in users to view their own previous evaluation runs.
MLE-F-027	The system shall allow logged-in users to delete their own evaluation-history entries.

5. Non-Functional Requirements

Each non-functional requirement is measurable and shall be verified through the test plan.

ID	Category	Requirement
MLE-NF-001	Performance	For a dataset of up to 10,000 rows and 50 columns, a single-algorithm evaluation shall return results within 15 seconds under the defined test environment.
MLE-NF-002	Usability	A first-time user shall be able to complete the upload-to-results flow without external instructions within 5 minutes during usability testing.
MLE-NF-003	Scalability	The backend shall support at least 20 concurrent evaluation requests without failure; requests beyond the supported level may be queued.
MLE-NF-004	Reliability	Malformed datasets and unsupported configurations shall produce clear, non-crashing error responses in at least 99% of tested invalid-input cases.
MLE-NF-005	Portability	The application shall function correctly on the latest two versions of Chrome, Firefox, and Edge.
MLE-NF-006	Maintainability	The evaluation engine shall be modular so that adding a supported algorithm requires changes confined to the relevant algorithm module/interface.

5.1 Usability and Accessibility

- The primary workflow shall be understandable without external instructions.
- Forms shall provide clear validation messages and actionable errors.
- Charts and tables shall use readable labels and sufficient visual contrast.

5.2 Reliability

- Invalid files and unsupported configurations shall be handled without server crashes.
- Evaluation results shall be associated with the correct user and evaluation run.

5.3 Portability

- The application shall support the latest two versions of Chrome, Firefox, and Edge.

6. Security Requirements

6.1 Security Objectives

- Protect user credentials and account information from unauthorized access.
- Prevent unauthorized access to uploaded datasets and evaluation history.
- Validate uploaded input and protect server-side processing.

6.2 Security Requirements

ID	Requirement
MLE-SEC-001	Passwords shall be stored using a strong one-way hashing algorithm such as bcrypt with a unique salt per user.
MLE-SEC-002	Uploaded files shall be validated and sanitized; unsupported formats, malformed input, and oversized files shall be rejected.
MLE-SEC-003	Authentication shall be required before accessing user-specific evaluation history.
MLE-SEC-004	HTTPS/TLS shall be enforced for all deployed client-server communication.
MLE-SEC-005	Authentication endpoints shall implement rate limiting to reduce brute-force attempts.

7. Quality Attributes and Acceptance Criteria

7.1 Quality Attributes

Attribute	Expected Quality
Performance	Evaluation completes within the specified response-time requirement for the reference dataset.
Usability	A first-time user can complete the primary workflow without external instructions.
Reliability	Invalid input produces controlled errors rather than application crashes.
Security	Credentials, sessions, datasets, and history are protected against unauthorized access.
Portability	Supported browsers provide consistent core functionality.
Maintainability	Algorithm integration is modular and isolated from unrelated components.

7.2 Acceptance Criteria

- All high-priority functional requirements are implemented and verified.
- No critical non-functional or security requirement fails.
- Required test cases are linked to requirements in the RTM.
- Authentication, dataset validation, model evaluation, visualization, export, history, security, and browser-compatibility test suites pass.

7.3 Acceptance Test Suites

- Authentication
- Dataset Upload and Validation
- Algorithm Evaluation
- Results Visualization
- Report Export
- History Management
- Security
- Cross-Browser Compatibility

8. System Models and Diagrams

8.1 UML Use-Case Model

The SRS requires two use-case views. The diagrams should be maintained separately from this requirements document and referenced here.

Diagram	Actors / Use Cases
Diagram 1 — User Workflow	Guest / Registered User: Register, Login, Upload Dataset, Configure Target and Features, Select Algorithm(s), Run Evaluation, View Results, Export Report, View History, Delete History Entry.
Diagram 2 — Administration (Optional)	Admin: View System Usage Statistics, View Logs.

8.2 Primary System Flow

Upload Dataset → Validate & Preview → Configure Target/Features → Select Algorithms & Split → Train & Evaluate → Display Metrics/Charts → Export Results → Save History (Registered Users)

9. Requirements Traceability Matrix (RTM)

The RTM links selected requirements to their intended design component and verification test case. Status is initially marked Pending until implementation and testing are completed.

Req ID	Requirement	Design Element	Test Case	Status
MLE-F-001	User registration	Auth module	TC-001	Pending
MLE-F-002	Password hashing	Auth module	TC-002	Pending
MLE-F-003	User login	Auth module	TC-003	Pending
MLE-F-006	Dataset upload	Upload module	TC-006	Pending
MLE-F-007	File validation	Upload module	TC-007	Pending
MLE-F-010	Target selection	Config module	TC-010	Pending
MLE-F-013	Algorithm list	Evaluation engine	TC-013	Pending
MLE-F-016	Model training & evaluation	Evaluation engine	TC-016	Pending
MLE-F-017	Metric computation	Evaluation engine	TC-017	Pending
MLE-F-020	Metric comparison chart	Frontend dashboard	TC-020	Pending
MLE-F-023	PDF export	Report module	TC-023	Pending
MLE-F-025	History save	History module	TC-025	Pending
MLE-NF-001	Performance	Evaluation engine	TC-N01	Pending
MLE-NF-005	Cross-browser support	Frontend	TC-N05	Pending
MLE-SEC-001	Password hashing	Auth module	TC-SEC-001	Pending
MLE-SEC-002	Upload validation	Upload module	TC-SEC-002	Pending
MLE-SEC-003	History access control	Auth module	TC-SEC-003	Pending
MLE-SEC-004	HTTPS/TLS	Infrastructure	TC-SEC-004	Pending
MLE-SEC-005	Login rate limiting	Auth module	TC-SEC-005	Pending

End of Document